

MIDTERM EXAM

ECO403/503—INTRODUCTION TO CONTRACT THEORY & AUCTIONS

FEBRUARY 23, 2025

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Problem 1: The NUpiterian Puzzle

In our galaxy, far beyond the reaches of Earth, lies the peculiar planet of **NUpiter**, home to the highly logical and tradition-bound **NUpiterians**. NUpiter is inhabited by K NUpiterians. These beings pride themselves on recognizing intellect and work ethic in others, yet, due to an ancient and unshakable belief system, *no NUpiterian may ever know whether they themselves are smart and hardworking!* Can you believe it?

Each NUpiterian can easily recognize intelligence and hard work in others, but their strange culture dictates that no NUpiterian can recognize intelligence and hard work within themselves. If a NUpiterian ever realizes that they are smart and hardworking, their religion compels them to perform a very scary ritual at noon on the following day: they must ascend to another plane of existence—a realm so terrifying that it is only spoken of in hushed whispers: “*Real Life*.”

Once in *Real Life*, a NUpiterian faces horrors beyond imagination: responsibilities, decision-making, and worst of all, getting a job. For these reasons, NUpiterian tradition dictates that they must live in carefully maintained ignorance, ensuring that no one ever meets this fate.

All NUpiterians are highly logical and devout in their customs, and they all know that each other is also highly logical and devout (and they all know that they all know that each other is highly logical and devout, and so forth).

Among the K NUpiterians, it turns out that N of them are smart and hardworking, where $N \leq K$, and the remaining $K - N$ are not. However, the NUpiterians are not initially aware of this distribution (each of them, of course, recognizes the intelligence and work ethic of the other $K - 1$ NUpiterians, but they do not know their own status).

One fateful day, a *dashing and daring explorer*, Prof. Alejandro, arrives on NUpiter. A fearless traveler with an *unparalleled knack for chaos*, he is instantly recognized as smart and hardworking by the locals, so he quickly gains their trust and is invited to deliver an address to the entire civilization.

Aware of their cultural taboos, but always with a taste for chaos, Prof. Alejandro mischievously remarks:

Ah, it’s wonderful to come in contact with other smart and hardworking beings such as myself!
I was starting to think I might be the only one in this region of the galaxy!

The moment these words leave his mouth, a horrified silence falls over the crowd. The NUpiterians, bound by logic, begin to think.

What effect, if anything, does this chaotic remark by Prof. Alejandro have on the NUpiterians?

- 1.1. Suppose first that $N = K = 3$. Let S denote “hardworking and studious”, and $\neg S$ denote “Not hardworking or studious”. Using the previous notation, represent this situation as a knowledge space with $2^3 = 8$ states of the world.
- 1.2. Use the knowledge space framework to prove that after three days, all three NUpiterians ascend to *Real Life*.
- 1.3. What new information did Prof. Alejandro introduce by making his announcement?
- 1.4. Generalize this result for arbitrary values of N and K where $N \leq K$.

Problem 2: NU Roommates

You (player 1) have a roommate (player 2). Your NU dormitory room is dirty. Cleaning it requires an effort cost of $c > 0$. Will either of you clean it? Each of you has two available actions: Clean (C) or go to the Movies (M).

Each player has two possible types, representing their valuation for a clean room: v_H (high value) and $v_L = 0$ (low value). The probability of being the high type is $p \in (0, 1)$.

Assume the following conditions on the parameters:

$$v_H > c \quad \text{and} \quad v_H(1 - p) < c.$$

The payoffs of a player with valuation v are given by:

$$u_i(a_i, a_j, v) = \begin{cases} v - c & \text{if } a_i = C, a_j \in \{C, M\}, \\ v & \text{if } a_i = M, a_j = C, \\ 0 & \text{if } a_i = a_j = M. \end{cases}$$

Remark: The payoff is the same whether both of you clean or only one of you does. If either player cleans, they bear the full cost c .

- 2.1. Describe the belief space of this game.
- 2.2. Show that a low-type player ($v_L = 0$) never cleans in equilibrium.
- 2.3. Let $\alpha_i = \sigma_i(C | v_H)$ be the behavioral strategy of player i when they have type v_H . Compute the expected payoffs of player i from cleaning and from going to the movies, given that the other roommate follows strategy α_j .
- 2.4. Provide a full description of each player's best response correspondence.
- 2.5. Compute all pure-strategy equilibria.
- 2.6. Compute all mixed-strategy equilibria.
- 2.7. Show that in any equilibrium, there exists a positive probability that at least one roommate values a clean room but does not clean it. This implies that the social outcome is not always efficient!
- 2.8. Suppose now that the payoffs for a player with valuation v are:

$$u_i(a_i, a_j, v) = \begin{cases} v - c/2 & \text{if } a_i = a_j = C, \\ v - c & \text{if } a_i = C, a_j = M, \\ v & \text{if } a_i = M, a_j = C, \\ 0 & \text{if } a_i = a_j = M. \end{cases}$$

- (a) Interpret these new payoffs and explain their meaning.
- (b) Repeat the analysis to find all equilibria.
- (c) Compare the results with the original game. What do you conclude?

Problem 3: Sharing information

Suppose that there are two players, 1, and 2. The set of states is $\Omega = \{1, 2, \dots, 12\}$ and the common prior of the two players is $\mu(\omega)$. Assume $\mu(\omega) > 0$ for all $\omega \in \Omega$.

Each player i observes, privately, the realization of the random variable $X_i : \Omega \rightarrow \mathbb{R}$, where these random variables are defined as follows:

$$X_1(\omega) = \begin{cases} 1 & \text{if } \omega \in \{1, 2, 3, 4\} \\ 2 & \text{if } \omega \in \{5, 6, 7, 8\} \\ 3 & \text{if } \omega \in \{9, 10, 11, 12\} \end{cases} \quad X_2(\omega) = \begin{cases} 1 & \text{if } \omega \in \{1, 2\} \\ 2 & \text{if } \omega \in \{3, 4\} \\ 3 & \text{if } \omega \in \{5, 6\} \\ 4 & \text{if } \omega \in \{7, 8\} \\ 5 & \text{if } \omega \in \{9, 10\} \\ 6 & \text{if } \omega \in \{11, 12\} \end{cases}$$

Let $A \in 2^\Omega$ be an event. Define $Q_i^0(\omega) = E[\mathbb{1}_A | X_i](\omega)$ for every $\omega \in \Omega$, where $\mathbb{1}_A$ is the indicator function for event A . Then, for $t = 1, 2, \dots$, define, for every $\omega \in \Omega$:

$$Q_1^t(\omega) = E[\mathbb{1}_A | X_1, Q_2^0, \dots, Q_2^{t-1}](\omega), \quad Q_2^t(\omega) = E[\mathbb{1}_A | X_2, Q_1^0, \dots, Q_1^{t-1}](\omega).$$

Answer the following True/False questions. For the answer to be valid, you should provide an argument/proof. Hint: All proofs here are very short.

- 3.1. True or False. For any μ and any event A , $\lim_{t \rightarrow \infty} Q_i^t$ exists for $i = 1, 2$ and the limits are equal.
- 3.2. True or False. For any μ and any event A , the players always end up with the same information partitions at the end of the process.

Problem 4: A proverb about knowledge

Consider the following Arabian proverb:

There are four sorts of men:

He who knows not and knows not he knows not: he is a fool—shun him;

He who knows not and knows he knows not: he is simple—teach him;

He who knows and knows not he knows: he is asleep—wake him;

He who knows and knows he knows: he is wise—follow him.

- 4.1. Let Ω be a finite set of states of the world. Let $N = \{F, S, A, W\}$ denote the set of agents, where F denotes the fool, S the simple, A the asleep, and W wise. Suppose that for each agent, we have a knowledge operator $L_i : 2^\Omega \rightarrow 2^\Omega$ that follows the setup of the proverb. For each agent $i \in N$, which one of Kripke's axioms do they violate?
- 4.2. Can any of the knowledge operators from each of the previous agents come from a standard model with information partitions?
- 4.3. Define a *possibility correspondence* as a function $P : \Omega \rightarrow 2^\Omega$ that is not necessarily partitional. Define its induced knowledge operator as usual

$$KA = \{\omega \in \Omega : P(\omega) \subseteq A\}.$$

Consider $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ and the following possibility correspondence:

$$P(\omega_1) = \{\omega_1, \omega_2\}$$

$$P(\omega_2) = \{\omega_2, \omega_3\}$$

$$P(\omega_3) = \{\omega_3\}$$

$$P(\omega_4) = \{\omega_4\}$$

Consider an agent whose knowledge operator K is induced by the previous possibility correspondence. Can this agent be the fool? Can this agent be the simple? Can this agent be asleep? Can this agent be wise?